MODULE 03

Regulatory Reports & Data Interpretation

eDNA Bioinformatics Learning Series · Yaw Baah

Modules 01 and 02 covered how the pipeline identifies species from sequences. This module focuses on what happens with those identifications: how to **interpret the results honestly**, how to **build a report a regulator will trust**, and what the numbers — Shannon H', Pielou J', confidence flags — actually mean for a real environmental survey.

What you will learn

1. What a regulatory eDNA report must contain 2. How to read a confidence flag 3. Shannon H' and Pielou J' explained in plain English 4. How to generate your report with --report 5. The four sections of the PDF report 6. Common interpretation mistakes 7. Glossary

1. Why Regulatory Reports Matter

Environmental DNA surveys are increasingly accepted by UK regulators as evidence for species presence or absence — but only when the data is **traceable, reproducible, and interpreted within known confidence limits**. A results CSV file is not a report. A regulator needs to know:

- Which species were detected and at what confidence level?
- What database and what version was used to identify them?
- How diverse is the community — and how was that measured?
- Who ran the analysis and when?
- Were any sequences unclassified — and why does that matter?

The pipeline's PDF report answers all five questions automatically. It is designed around the evidence standards in the **GB Non-Native Species Secretariat** and **Environment Agency** eDNA guidance, and the **NERC-funded eDNA Metabarcoding Framework**.

Who uses eDNA reports?

- **Environment Agency** — water quality, invasive species surveillance
- **Natural England / NatureScot / NRW** — protected species surveys (great crested newt, white-clawed crayfish, water vole)
- **Planning consultants** — habitat assessments for development planning applications
- **Water companies** — reservoir and catchment biodiversity monitoring

2. Reading Confidence Flags

Every row in the species table carries a **confidence_flag**: **HIGH**, **MEDIUM**, or **LOW**. This is the single most important column for a regulator. Here is exactly how it is calculated and what it means:

Flag	Identity threshold	What the pipeline reports	How to communicate it
HIGH	≥ 97 %	Species name (e.g. *Chlorella vulgaris*)	State: '*C. vulgaris* detected with high confidence.' Safe to report at species level.
MEDIUM	90 – 96 %	Genus only (e.g. *Daphnia* sp.)	State: '*Daphnia* sp. detected at genus level — species uncertain.' Do **not** name a species.
LOW	< 90 %	Genus sp. or unclassified	State: 'Sequence detected but confidence too low for reliable taxonomic assignment.' Flag for review.

LCA and the confidence_flag work together

The **confidence_flag** is set from the top-hit identity. But the **resolved_species** column — what the report actually shows — also applies LCA logic across all hits. So a sequence may have a HIGH flag (top hit ≥ 97 %) but still be reported as 'Genus sp.' if multiple hits with near-identical scores disagree on the species. Both columns are in blast_hit_table.csv for full transparency.

3. Diversity Indices — Plain English

The pipeline calculates two biodiversity indices: **Shannon H'** and **Pielou J'**. These summarise the community into two numbers that ecologists and regulators recognise. Understanding them helps you explain your results in plain language.

Shannon diversity index (H')

Shannon H' asks: *how much uncertainty is there about what species the next randomly-chosen sequence will belong to?* High uncertainty = many species, all roughly equally abundant = high diversity.

Formula: **H' = –Σ p■ · ln(p■)** where p■ is the proportion of reads belonging to species i.

H' value	Typical interpretation
0.0	Only one taxon present (H' = 0 when there is no uncertainty)
0.5 – 1.5	Low diversity — one taxon strongly dominates
1.5 – 2.5	Moderate diversity — several taxa, uneven
2.5 – 3.5	High diversity — many taxa, relatively even
> 3.5	Very high — typical of pristine, species-rich habitats

Pielou evenness index (J')

Pielou J' answers: *are the taxa evenly distributed, or does one dominate?* It is always between 0 and 1. It is calculated as **J' = H' / ln(S)** where S is species richness. This normalises Shannon's index so you can compare communities with different numbers of species.

J' value	What it means
1.0	Perfectly even — every taxon has exactly the same read count
0.8 – 0.99	Highly even — no single taxon dominates
0.5 – 0.79	Moderately uneven — one or two taxa more abundant
< 0.5	Strongly dominated — one taxon accounts for most reads

Worked example — Cannock Chase Pond A (demo run)

Species richness S = 6 | Shannon H' = 1.748 | Pielou J' = 0.976

Interpretation: Six taxa identified. The community is **highly even** (J' = 0.976 — nearly perfect evenness) with **moderate overall diversity** (H' = 1.748). *Microcystis aeruginosa* appeared in two sequences (the cyanobacterial bloom) but because total richness is 6 and reads are spread across them, evenness remains high. One sequence (mixed diatom genera) returned 'unclassified' and is excluded from diversity calculations — this is transparently reported as 'Unclassified: 1'.

4. Generating Your Report

Add **--report** to any pipeline run and a PDF is written alongside the CSV files. Three extra flags fill in the cover page:

```
python pipeline.py --fasta data/sample_sequences.fasta --local --db-path
/data/blast/nt --threads 4 --report --site "Cannock Chase Pond A" --sample-date
2026-05-22 --analyst "Yaw Baah"
```

Flag	Effect
--report	Triggers PDF generation (off by default)
--site NAME	Site name printed on the cover page and methodology section
--sample-date DATE	Collection date (YYYY-MM-DD). Defaults to today if omitted
--analyst NAME	Analyst name on cover and methodology

The PDF is saved as **data/results/eDNA_Report.pdf** (or whatever --output directory you set). If --local was used, the database version from db_version.json is automatically embedded in the Methodology section.

5. The Four Sections of the PDF Report

Cover page	Site name, sample date, analyst, report date. Plain-English description of the method. Designed to orient a non-technical regulator before they reach the numbers.
1. Executive Summary	Five metric tiles (species richness, H', J', sequences analysed, unclassified). Auto-generated plain-English interpretation. Confidence level guide table showing what HIGH / MEDIUM / LOW mean.
2. Species Identification Table	One row per sequence (top hit only). Columns: resolved species, confidence flag (colour-coded pill), identity %, e-value, accession. LOW confidence rows are shaded red to draw the eye immediately.
3. Biodiversity Metrics	Alpha-diversity table (S, H', J') with formulas and interpretation. Taxon count table with inline bar chart showing relative abundance. Unclassified sequences are counted but excluded from diversity.
4. Methodology	Pipeline description, LCA logic, identity thresholds. Database version box (from db_version.json). Sample metadata table (site, date, analyst, report date). Reproducibility statement.

6. Common Interpretation Mistakes

Mistake: Reporting a MEDIUM hit at species level

A 93 % identity hit to *Daphnia pulex* does NOT mean *D. pulex* was present. The pipeline correctly reports '*Daphnia* sp.' — stick to that in your write-up.

Mistake: Treating 'unclassified' as a missing data problem

Unclassified sequences are not errors — they are real sequences the database cannot confidently place. They may represent novel taxa, database gaps, or degraded DNA. Always report the unclassified count.

Mistake: Comparing H' across datasets with different --max-hits

Shannon H' in this pipeline is based on top-hit read counts. If one run used --max-hits 5 and another used --top-hit-only, the diversity numbers are not directly comparable.

Mistake: Ignoring LOW confidence rows

LOW confidence hits are shown in red in the report. Do not silently include them in a species list — flag them explicitly. They may indicate PCR noise, chimeric sequences, or genuine rarities.

Mistake: Not recording the database version

If you use web BLAST and re-run six months later, you may get different species names — NCBI updates daily. Always use --local with a version-stamped database for any analysis that will be submitted to a regulator.

7. Glossary

confidence_flag
Column in blast_hit_table.csv. 'high' (≥97 %), 'medium' (90–96 %), or 'low' (<90 %) based on the identity_pct of the top BLAST hit.

resolved_species
The post-LCA species assignment used for reporting and diversity calculations. May differ from the raw 'species' column if LCA logic downgraded the rank.

species_richness (S)
Count of distinct classified taxa in the top-hit table. Unclassified sequences are excluded.

Shannon index (H')
H' = -Σ p■ ln(p■). A measure of both the number of species and their relative abundances. Higher = more diverse. Maximum value = ln(S) (when all species are equally abundant).

Pielou evenness (J')
J' = H' / ln(S). Normalised evenness between 0 and 1. 1.0 = all taxa equally abundant. 0.0 = one taxon accounts for all reads (impossible in practice).

unclassified_queries
Sequences where resolved_species = 'unclassified'. Counted separately in the report — not included in richness or H'.

LCA (Lowest Common Ancestor)
A strategy for resolving disagreeing BLAST hits. If hits to a query all belong to the same genus but different species, the assignment is downgraded to 'Genus sp.' This avoids falsely naming a species when the evidence is ambiguous.

blast_hit_table.csv
The primary output file. One row per BLAST hit per query. Contains: query_id, species, resolved_species, confidence_flag, identity_pct, evalue, bit_score, query_coverage_pct, accession.

biodiversity_summary.csv
Secondary output. One metric per row — S, H', J', total_queries, unclassified_queries, plus one row per taxon with read count.

db_version.json
Written on every local BLAST run. Records the database path, version string, program, and UTC timestamp. Embedded in the PDF Methodology section automatically.

Key Takeaways

• A results CSV is not a regulatory report — use **--report** to generate a traceable, client-ready PDF.
• **Confidence flags are not optional context** — they are the primary quality signal. Never report a MEDIUM or LOW hit at species level.
• **Shannon H'** measures diversity; **Pielou J'** measures evenness. Report both — they tell different stories.
• **Unclassified sequences** must be reported, not hidden. They are part of the evidence base.
• **Database version** must be recorded for regulatory submissions. The pipeline does this automatically via db_version.json.

Next: Module 04 — Protected Species Detection & UK Regulatory Frameworks